School Worksheet

# Class 12 Mathematics, Chapter 13: Probability

Maximum marks: **51** **20** questions

Name _____ Date _____ Score _____ / 51

1. All questions are compulsory. Marks are shown against each question.
2. Show every step of working: steps carry marks, not just the final answer.
3. Where a question offers OR, attempt only one of the two choices.

| Q No. | Question | Marks |
|---|---|---|

## Level 1

8 marks

**1** MCQ — 1 mark

If $P(A) = 0.8$, $P(B) = 0.5$ and $P(B|A) = 0.4$, then $P(A \cap B)$ is:

**(A)** $0.32$ **(B)** $0.25$ **(C)** $0.1$ **(D)** $0.4$

**2** MCQ — 1 mark

The probability that a particular item is available in three shops A, B and C is $\frac{4}{5}$, $\frac{3}{4}$ and $\frac{2}{3}$ respectively. If a person visits all three shops, the probability that the item is available in at least one of them is:

**(A)** $\frac{59}{60}$ **(B)** $1$ **(C)** $\frac{1}{60}$ **(D)** $\frac{9}{60}$

**3** MCQ — 1 mark

A die is thrown. If $E$ is the event "the number is 4" and $F$ is the event "the number is even", then $P(E|F)$ is:

**(A)** $\frac{1}{6}$ **(B)** $\frac{1}{2}$ **(C)** $\frac{1}{3}$ **(D)** $\frac{2}{3}$

**4** MCQ — 1 mark

If $P(A) = \frac{1}{2}$ and $P(B) = 0$, then $P(A|B)$ is:

**(A)** $0$ **(B)** $\frac{1}{2}$ **(C)** $1$ **(D)** not defined

**5** MCQ — 1 mark

A random variable $X$ has the distribution $P(X = 0) = k$, $P(X = 1) = 2k$, $P(X = 2) = 3k$. The value of $k$ is:

**(A)** $\frac{1}{6}$ **(B)** $\frac{1}{3}$ **(C)** $\frac{1}{2}$ **(D)** $1$

| Q No. | Question | Marks |
|---|---|---|
| 6 | Assertion-Reason<br>**Assertion (A):** Events $A$ and $B$ with $P(A)=\frac{1}{2}$, $P(B)=\frac{1}{3}$ and $P(A\cap B)=\frac{1}{6}$ are independent. **Reason (R):** Two events $A$ and $B$ are independent if and only if $P(A\cap B)=P(A)\,P(B)$.<br>**(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).<br>**(B)** Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A).<br>**(C)** Assertion (A) is true, but Reason (R) is false.<br>**(D)** Assertion (A) is false, but Reason (R) is true. | 1 mark |
| 7 | Short answer<br>Two cards are drawn one after the other without replacement from a well-shuffled pack of 52 cards. Find the probability that both are kings. | 2 marks |

## Level 2 — 18 marks

| | | |
|---|---|---|
| 8 | Short answer<br>The chances of P, Q and R getting selected as CEO of a company are in the ratio $4:1:2$. The probabilities that the company increases its profits under the new CEO P, Q or R are $0.3$, $0.8$ and $0.5$ respectively. If the company increased its profits, find the probability that it is due to the appointment of R as CEO. | 3 marks |
| 9 | Short answer<br>Find the probability distribution of the number of heads in two tosses of a fair coin. Also find its mean. | 2 marks |
| 10 | Short answer<br>Given $P(A)=0.6$, $P(B)=0.3$ and $P(A\cap B)=0.2$, find $P(A\|B)$, $P(B\|A)$ and $P(A'\|B')$. | 3 marks |
| 11 | Short answer<br>A and B independently try to solve a problem with probabilities $\frac{1}{2}$ and $\frac{1}{3}$ respectively. Find the probability that (a) the problem is solved (b) exactly one of them solves it. | 3 marks |
| 12 | Short answer<br>A family has two children. Find the probability that both are boys, given that at least one of them is a boy. | 2 marks |
| 13 | Short answer<br>A random variable $X$ has $P(X=x)=kx$ for $x=1,2,3,4$ and $0$ otherwise. Find $k$, $P(X\geq 3)$ and the mean of $X$. | 3 marks |
| 14 | Short answer<br>Three coins are tossed. Let $E$: “three heads or three tails” and $F$: “at least two heads”. Are $E$ and $F$ independent? | 2 marks |

| Q No. | Question | Marks |
|---|---|---|
| **Level 3** | | **25 marks** |
| **15** | Long answer<br>A factory has machines A, B and C producing $50\%$, $30\%$ and $20\%$ of the items, of which $2\%$, $3\%$ and $4\%$ respectively are defective. An item picked at random is found defective. Find the probability that it was produced by machine A. | 5 marks |
| **16** | Short answer<br>In answering a multiple-choice question (4 options), a student either knows the answer or guesses. The probability that she knows the answer is $\frac{3}{4}$. If she answers correctly, find the probability that she knew the answer. | 3 marks |
| **17** | Long answer<br>Two cards are drawn successively without replacement from a well-shuffled pack of 52 cards. Find the probability distribution of the number of aces and its mean. | 5 marks |
| **18** | Short answer<br>A random variable $X$ has the distribution: $X$: $0, 1, 2, 3, 4, 5, 6, 7$ with $P(X)$: $0,\ k,\ 2k,\ 2k,\ 3k,\ k^2,\ 2k^2,\ 7k^2+k$. Find $k$, $P(X<3)$ and $P(X>6)$. | 3 marks |
| **19** | Long answer<br>A laboratory blood test is $99\%$ effective in detecting a disease when it is present, but gives a false positive for $0.5\%$ of healthy persons. If $0.1\%$ of the population has the disease, find the probability that a person who tests positive actually has the disease. | 5 marks |
| **20** | Case study<br>**Case Study:** An NGO organises a charity event to distribute woollen caps to children in winter. The caps are in three boxes: Box I has 30 red caps, Box II has 20 red and 10 green caps, and Box III has 30 green caps. The probability that Box $i$ is selected and a cap picked out is $\frac{i}{6}$, $i=1,2,3$. A person selects a cap.<br>**(i)** What is the probability that he selects a red cap? — 2 marks<br>**(ii)** If he selects a green cap, what is the probability that it has come from Box II? — 2 marks | 4 marks |

# Answer key

Chapter 13: Probability, School Worksheet. Full step-wise solutions are in the solutions sheet.

## Level 1

| | |
|---|---|
| 1 | (A) $0.32$ |
| 2 | (A) $\frac{59}{60}$ |
| 3 | (C) $\frac{1}{3}$ |
| 4 | (D) not defined |
| 5 | (A) $\frac{1}{6}$ |
| 6 | (A) Both A and R are true and R is the correct explanation of A |
| 7 | $\frac{1}{221}$ |

## Level 2

| | |
|---|---|
| 8 | $P(E_3 \mid A) = \frac{1}{3}$ |
| 9 | $X$: $0, 1, 2$ with $P(X)$: $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$; mean $= 1$ |
| 10 | $P(A \mid B) = \frac{2}{3}$, $P(B \mid A) = \frac{1}{3}$, $P(A' \mid B') = \frac{3}{7}$ |
| 11 | (a) $\frac{2}{3}$ (b) $\frac{1}{2}$ |
| 12 | $\frac{1}{3}$ |
| 13 | $k = \frac{1}{10}$, $P(X \geq 3) = \frac{7}{10}$, mean $= 3$ |
| 14 | Yes, $P(E \cap F) = \frac{1}{8} = P(E)\,P(F)$, so $E$ and $F$ are independent |

## Level 3

| | |
|---|---|
| 15 | $P(E_1 \mid D) = \frac{10}{27}$ |
| 16 | $P(E_1 \mid A) = \frac{12}{13}$ |
| 17 | $X$: $0, 1, 2$ with $P(X)$: $\frac{188}{221}, \frac{32}{221}, \frac{1}{221}$; mean $= \frac{2}{13}$ |
| 18 | $k = \frac{1}{10}$, $P(X < 3) = \frac{3}{10}$, $P(X > 6) = \frac{17}{100}$ |
| 19 | $P(E_1 \mid A) = \frac{22}{133}$ |
| 20 | (i) $\frac{7}{18}$ (ii) $\frac{2}{11}$ |